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Evaluation of a Rapid, Low-Cost Broth Turbidity Test for Detecting Ampicillin-Resistant Lower Urinary Tract Infections in Dogs and Cats

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ABSTRACT

Background: Effective treatment of urinary tract infections (UTIs) in dogs and cats relies on timely recognition of antimicrobial resistance, but traditional microbiological culture and susceptibility testing require 48–72 h and can be cost-prohibitive.

Objective: To evaluate a novel, low-cost broth turbidity test for detecting ampicillin-resistant UTIs in dogs and cats compared to the gold standard microbiological methods.

Animals: One hundred sixty urine samples from 145 dogs and 15 cats at the Mississippi State University College of Veterinary Medicine.

Methods: From September 2023 to February 2024, urine samples were tested using an ampicillin-supplemented Mueller–Hinton broth turbidity test and gold standard culture with Sensititre MIC plates. Sensitivity, specificity, Cohen's κ, and McNemar's test were calculated.

Results: The broth turbidity test showed 89.47% sensitivity (95% CI: 66.86%–98.72%) and 100% specificity (95% CI: 97.42%–100%) for the detection of ampicillin-resistant organisms in the urine, with substantial agreement ($\kappa = 0.94$, $p < 0.001$) and no classification difference ($p = 0.500$). Notably, all 36 samples with ampicillin-susceptible organisms identified by the gold standard cultures test were correctly identified as negative for ampicillin resistance by the broth turbidity test.

Conclusions: The broth turbidity test's simplicity and affordability make it a promising tool for in-house use by veterinary practitioners to guide first-line therapy decisions, though its sensitivity could be limited in cases with low bacterial loads or confounding factors such as recent antibiotic administration.

1 | Introduction

Urinary tract infections (UTIs) are prevalent in dog and cat populations, often requiring prompt antimicrobial therapy to prevent complications such as pyelonephritis or recurrent infections. Amoxicillin and potentiated sulfonamides are both considered first-line treatments for uncomplicated UTIs, but amoxicillin is preferred by many practitioners due to its efficacy, safety, and cost-effectiveness [1]. Amoxicillin is excreted into the urine at a high concentration, which likely reduces the need

Abbreviations: CI, confidence interval; MH, Mueller–Hinton; MIC, minimal inhibitory concentration; MSU-CVM, Mississippi State University College of Veterinary Medicine; UTI, urinary tract infection.

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for clavulanic acid for successful treatment of many infections [1]. However, the rising prevalence of ampicillin-resistant bacteria, particularly methicillin-resistant *Staphylococcus pseudintermedius* and extended-spectrum beta-lactamase-producing Enterobacterales, has complicated empirical therapy, leading to treatment failures and the use of second-line antibiotics [2]. Traditional microbiological culture and susceptibility testing, while the gold standard, are time-consuming (48–72 h) and usually require submission to off-site diagnostic laboratories [3]. Published veterinary guidelines state that "aerobic bacterial urine culture is preferred for all cases of suspected bacterial cystitis," but surveys suggest that practitioners commonly prescribe antibiotics empirically for these cases, with cost being a major limiting factor [1, 4–7]. Low-cost diagnostic tools are essential for veterinarians working in resource-limited settings and for those providing care across the spectrum of care model. These tools support decision-making by enabling diagnostic testing that aligns with clients' financial limitations while still promoting responsible antimicrobial use [8].

Rapid, cost-effective point-of-care diagnostic tools to detect ampicillin resistance in dog and cat UTIs are lacking, hindering timely and appropriate treatment decisions. Such tools are needed to guide first-line therapy, reduce unnecessary antibiotic escalation, and support antimicrobial stewardship.

The objective of this study was to develop a rapid, cost-effective, user-friendly microbial assay to reliably detect ampicillin-resistant bacteria in urine samples within 16–24 h as a point-of-care diagnostic. If the pathogen grows in the presence of the indicator drug (ampicillin), the broth will become visibly turbid, indicating positive growth of an amoxicillin-resistant microbe. A positive result will prompt the clinician to submit the sample for traditional culture and susceptibility testing at an accredited laboratory, thus avoiding inappropriate first-line therapy resulting in likely improved patient outcomes. This study aimed to evaluate the performance of the experimental broth turbidity test against traditional microbiological methods in detecting ampicillin-resistant lower UTIs in dogs and cats. We hypothesized that the test would exhibit high sensitivity and specificity for the detection of ampicillin-resistant bacteria in dog and cat urine, making it a practical tool for in-house use by veterinary practitioners.

2 | Materials and Methods

2.1 | Study Design and Sample Collection

This study was conducted in accordance with the Standards for Reporting of Diagnostic Accuracy Studies (STARD) guidelines to ensure transparent and comprehensive evaluation of the broth turbidity test's diagnostic accuracy [9]. Urine samples submitted for routine culture and antibiotic susceptibility testing (AST) at the Mississippi State University College of Veterinary Medicine (MSU-CVM) Diagnostic Laboratory from September 2023 to February 2024 were tested in parallel using the experimental broth turbidity test. A total of 160 de-identified urine samples collected via cystocentesis from 145 dogs and 15 cats were used, obtained as residual samples from routine diagnostic submissions unrelated to this study. The MSU-CVM Institutional Animal Care and Use Committee (IACUC) reviewed the study and determined that no IACUC protocol was required, as no live animals were involved and only pre-existing diagnostic samples were analyzed. Owner consent for diagnostic testing was obtained as part of standard clinical procedures at the MSU-CVM Diagnostic Laboratory.

Due to the absence of published data on the broth turbidity test's performance, a sample size of 160 urine samples was selected based on unpublished antibiogram data from Mississippi State University (2022–2023), estimating a 15% prevalence of ampicillin-resistant bacteria in dog and cat urine cultures. Assuming a 15% prevalence and at least a 90% sensitivity for the broth turbidity test (informed by our unpublished pilot experiments), approximately 24 resistant cases were expected, yielding a 95% confidence interval (CI) for sensitivity with a width of approximately ±10% using the Clopper–Pearson method [10]. This precision was deemed sufficient for evaluating diagnostic accuracy, consistent with STARD guidelines [9].

2.2 | Gold Standard Traditional Microbiological Methods

Gold standard testing followed standard protocols for urinary bacterial culture and AST, as described by the Clinical Laboratory Standards Institute (CLSI). In summary, a calibrated 1 μL sterile loop was used to inoculate blood agar, Columbia nalidixic acid (CNA) agar, MacConkey agar, and an enrichment broth with urine. After 24 h of aerobic incubation at 37°C, colonies were manually selected from agar plates, isolated, and identified using Sensititre gram positive or negative ID plates (Thermo Fisher Scientific). If agar plates failed to grow after 24 h but the enrichment broth exhibited growth, subcultures were performed for bacterial isolation. Ampicillin susceptibility was determined using Sensititre CMV1BURF minimum inhibitory concentration (MIC) plates, with results interpreted per CLSI guidelines. An organism was classified as resistant (positive) if its MIC for ampicillin exceeded the CLSI susceptible breakpoint for lower UTIs in dogs and cats (8 μg/mL) [11].

2.3 | Experimental Broth Turbidity Test

The broth turbidity test was designed for simplicity and affordability. A clear sterile polystyrene tube containing 2.5 mL of sterile cation-adjusted Mueller–Hinton (MH) broth and two 10 μg ampicillin discs (corresponding to the CLSI breakpoint of 8 μg/mL) were added. The tube was vortexed to ensure disc dissolution, then inoculated with 20 μL of urine (two 10 μL sterile loops). Twenty-microliters of urine was used as the inoculation volume based on our unpublished pilot studies that suggested it would be a reasonable volume. Broth tubes were incubated aerobically in a non-shaking incubator at 37°C overnight for 16–24 h. After incubation, tubes were gently hand-rocked to suspend any bacteria and visually assessed by two independent and blinded observers for turbidity compared to a negative control tube, with apparent turbidity indicating growth of ampicillin-resistant bacteria (positive), as demonstrated in Figure 1. Tubes with no visible turbidity were classified as no growth of ampicillin-resistant bacteria (negative). Observers were blinded to the results of the

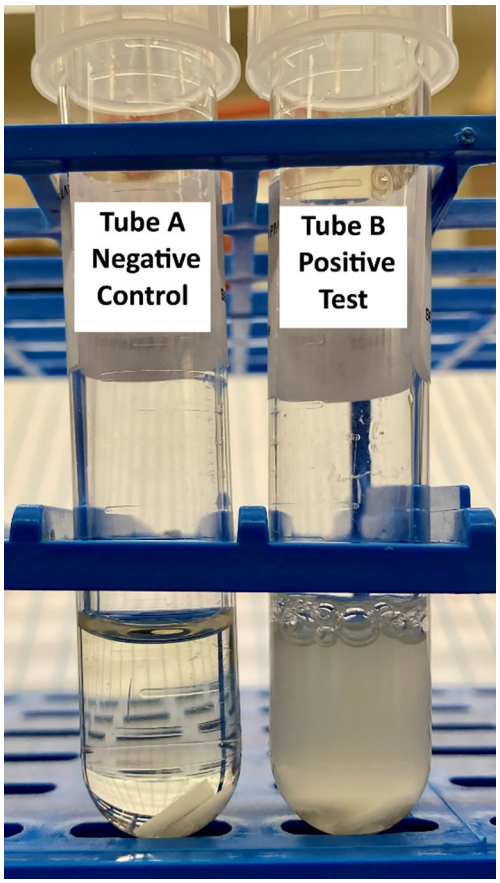

FIGURE 1 | Broth turbidity test visual comparison. Tube A = negative control tube; Tube B = positive sample.

gold standard test at the time of turbidity assessment; gold standard results were not available until 24–48 h after turbidity was assessed.

2.4 | Statistical Analysis

The performance of the broth turbidity test was compared to the gold standard for detecting ampicillin-resistant bacteria. A 2×2 contingency table was constructed to calculate true positives, true negatives, false positives, and false negatives. Sensitivity and specificity were computed, with 95% CIs determined using the exact binomial (Clopper–Pearson) method via an online calculator [10]. Cohen's κ was calculated to assess agreement beyond chance, and McNemar's test was used to evaluate differences in classification between the two methods. Statistical analyses were performed using SPSS (version 28, IBM Corp.) or the Soper online calculator, with significance set at $p < 0.05$.

3 | Results

3.1 | Summary of Bacterial Species and Ampicillin MICs

Urine from 145 dogs and 15 cats submitted for routine AST was used for the study. Of the 160 urine samples analyzed, the gold standard method identified bacterial growth in 55 samples (34.4%), while 105 samples (65.6%) showed no growth. Among the 55 samples with bacterial growth, 36 (65.5%) contained only ampicillin-susceptible isolates (MIC ≤ 8 μg/mL), and 19 (34.5%) had at least one ampicillin-resistant isolate (MIC > 8 μg/mL). The experimental broth turbidity test, designed to detect ampicillin-resistant organisms, showed growth in 17 samples (10.6%) and no growth in 143 samples (89.4%). The broth test correctly classified all 36 ampicillin-susceptible samples as negative for ampicillin-resistant organisms (no growth) but misclassified 2 of the 19 gold standard identified ampicillin-resistant samples as no growth (negative for ampicillin resistance), indicating false negatives. Interobserver agreement for the broth turbidity test was 100%, with both observers classifying all samples the same (negative vs. positive). A 2×2 contingency table summarizing the detection of resistant organisms is presented in Table 1. The broth turbidity test demonstrated a sensitivity of 89.47% (95% CI: 66.86%–98.72%) and a specificity of 100% (95% CI: 97.42%–100%) for detecting ampicillin-resistant organisms. Cohen's κ was 0.94 ($p < 0.001$), indicating almost perfect agreement between the broth turbidity test and the gold standard. McNemar's test showed no significant difference in classification between the two methods ($p = 0.500$, exact test, two-sided).

TABLE 1 | Contingency table for the broth turbidity test versus gold standard test.

Gold standard	Broth turbidity positive	Broth turbidity negative	Total
Pos	17 (True positives)	2 (False negatives)	19
Neg	0 (False positives)	141 (True negatives)	141
Total	17	143	160

Susceptible isolates included a variety of species, predominantly *Escherichia coli* ($n = 10$) and *Enterococcus faecalis* ($n = 7$), with MICs ranging from < 0.12 to 4 μg/mL. Of the 36 ampicillin-susceptible cultures, 6 contained two isolates each. Resistant isolates were primarily *E. coli* ($n = 11$), with MICs ranging from 32 to > 256 μg/mL, alongside other species such as *Klebsiella pneumoniae* and *Enterobacter cloacae*. Of the 19 ampicillin-resistant cultures, 6 contained two isolates. Three of these cultures contained two morphologically distinct *E. coli* isolates, which were identified based on differences in colony morphology noted during isolation. Two discrepant result cultures were noted: one contained a mixed culture of *S. pseudintermedius* (MIC < 2 μg/mL) and *Proteus* spp. (MIC 16 μg/mL), and the other contained a β-lactamase-producing *E. coli* isolate (MIC > 256 μg/mL). Both discrepant result samples tested negative in the broth turbidity test despite demonstrating resistance by the gold standard. Table 2 summarizes the bacterial species and MIC ranges, categorized by susceptibility, resistance, and discrepant results.

Notably, all 36 cultures identified by the gold standard as positive for ampicillin-susceptible organisms (e.g., *Streptococcus* spp., *Staphylococcus pseudintermedius*, *E. coli*) were correctly classified as negative for ampicillin resistance by the broth turbidity test. This further confirms the test's 100% specificity for

TABLE 2 | Summary of bacterial species and ampicillin MICs.

Category	Species	Number of isolates	MIC range (μg/mL)
Susceptible	*Streptococcus canis*	3	< 2 to ≤ 2
Susceptible	*Enterococcus faecalis*	7	< 2 to ≤ 2
Susceptible	Coagulase-negative *Staphylococcus* spp.	1	< 0.12
Susceptible	*S. warneri*	1	< 2
Susceptible	*S. aureus*	1	< 2
Susceptible	*Escherichia coli*	10	≤ 2 to 4
Susceptible	*Staphylococcus pseudintermedius*	5	< −0.25 to < 2
Susceptible	*Proteus* spp.	5	< 2 to ≤ 2
Susceptible	*Bacillus* spp.	2	< 0.25 to < 2
Susceptible	*Staphylococcus epidermidis*	1	2
Resistant	*E. coli*	11	32 to > 256
Resistant	*E. faecalis*	2	> 8
Resistant	*Klebsiella pneumoniae*	2	32 to > 256
Resistant	*S. pseudintermedius*	2	16–64
Resistant	*Enterobacter cloacae*	2	> 8 to > 256
Resistant	*Pseudomonas aeruginosa*	1	> 256
Resistant	*Proteus* spp.	1	16
Discrepant Result 1[a]	*Proteus* spp.[a]	1	16
Discrepant Result 1[a]	*Staphylococcus pseudintermedius*[a]	1	< 2
Discrepant Result 2	*E. coli*	1	> 256

[a]Discrepant Result 1 had two bacterial isolates: Ampicillin-susceptible *S. pseudintermedius* and ampicillin-resistant *Proteus* spp.

detection of ampicillin resistance (zero false positives), even in the presence of uropathogenic bacteria.

The two discrepant results (false negatives) involved samples with low bacterial loads. One sample (Discrepant Result 1) had 1000 CFUs/mL in the gold standard test and contained ampicillin-susceptible *S. pseudintermedius* and ampicillin-resistant *Proteus* spp., while the other (Discrepant Result 2) had < 1000 CFUs/mL, grew only in the gold standard's enrichment broth, and contained ampicillin-resistant *E. coli*. These discrepant results were included in the analysis to ensure a conservative estimate of the test's sensitivity. Further investigation revealed that Discrepant Result 1 was from an animal treated with cefovecin 2 days before sample collection, likely resulting in high urinary concentrations of cefovecin at the time of culture [12]. This almost certainly contributed to suppresion of bacterial growth in the broth turbidity test, rendering the result invalid [13]. Discrepant Result 2 involved a *Proteus* species with an ampicillin MIC of 16 μg/mL on the Sensititre MIC plate, which is within one dilution of the CLSI breakpoint (8 μg/mL). Due to the design of MIC plates (antibiotic concentrations double with each well), a reported MIC of 16 μg/mL actually indicates that the true MIC is > 8 but ≤ 16 μg/mL.

4 | Discussion

This study evaluated a novel, low-cost broth turbidity test for detecting ampicillin-resistant lower UTIs in dogs and cats, with the goal of guiding first-line therapy decisions. The test exhibited high specificity (100%) and substantial agreement with the gold standard ($\kappa = 0.94$), making it a reliable tool for ruling out ampicillin resistance. Its sensitivity (89.47%) was slightly lower, primarily due to two false negatives in samples with low bacterial loads (≤ 1000 CFUs/mL). These discrepant results were retained in the analysis to provide a conservative estimate of sensitivity, ensuring that the test's performance is not overestimated. The findings suggest that the test is a promising in-house option for veterinary practitioners, particularly in resource-limited settings where rapid results can inform whether amoxicillin is appropriate or if a formal culture is warranted.

The test's 100% specificity is a key strength, as demonstrated by its ability to correctly classify all 36 cultures positive for ampicillin-susceptible organisms as negative for resistance. This ensures that practitioners can confidently use amoxicillin in such cases without risking inappropriate treatment escalation. The high positive predictive value in this sample

population (100%) further supports its utility in identifying true resistant cases, prompting submission for formal culture and susceptibility testing. However, an important limitation of the broth turbidity test is that it is specifically designed to detect ampicillin-resistant organisms and does not distinguish between antibiotic-susceptible infections and cultures with no bacterial growth. As a result, a negative test result could indicate either the absence of infection or the presence of a susceptible infection, but the test cannot differentiate between these scenarios. Therefore, this test should only be used when the clinician has a high clinical suspicion of a lower UTI that warrants empiric treatment. If the turbidity test on a lower UTI urine sample suggests ampicillin susceptibility, clinicians can treat with confidence using guideline-recommended amoxicillin. If the turbidity test suggests amoxicillin resistance, amoxicillin could still be an effective treatment, but results will be less predictable. In this case, the client could be given the option to opt for a different empirical drug (e.g., potentiated sulfonamide) based on local antibiogram data or submit the urine sample for traditional bacterial culture and AST at a diagnostic lab.

The two false negatives highlight additional limitations. Discrepant Result 1 was likely confounded by recent cefovecin administration, which could have suppressed bacterial growth in the broth turbidity test due to high urinary concentrations of the antibiotic. Cefovecin, a third-generation cephalosporin, has a long half-life in dogs and cats, and it is well documented that antibiotics excreted into urine at the time of bacterial culture can inhibit the growth of organisms [12, 13]. In standard AST protocols, organisms are first isolated and grown in antibiotic-free media, then tested for susceptibility using purified colonies [14]. This approach allows bacteria to undergo exponential growth under ideal conditions before antibiotic exposure. In contrast, the broth turbidity test exposes bacteria to ampicillin without a prior isolation step. So, the bacteria were effectively transferred from cefovecin-containing urine directly to ampicillin-containing MH broth, potentially compounding the inhibitory effects of the antibiotics. These findings suggest that the broth turbidity test might be unreliable in patients recently treated with antibiotics and that clinical history, particularly recent antimicrobial use, should be carefully considered when interpreting results.

Discrepant Result 2 involved a *Proteus* species with an MIC of 16 μg/mL, just one dilution above the CLSI breakpoint of 8 μg/mL for ampicillin resistance and the concentration of ampicillin in the broth turbidity test. CLSI guidelines state that a margin of error of ±1 dilution is acceptable for MIC testing [14]. Additionally, reported MIC values are the highest value within the dilution range, meaning a reported MIC value of 16 has a true MIC value > 8 but ≤ 16 μg/mL [3]. Variability in the antibiotic content of Kirby–Bauer discs is also well documented, with studies reporting high variability in the true antibiotic content of discs [15]. If the true MIC of the *Proteus* isolate was closer to 8.1 μg/mL and the disc content was slightly above 10 μg (e.g., 11 μg), this could explain why the organism did not grow in the broth turbidity test, leading to a false negative. This highlights the importance of considering the inherent variability in both MIC testing and disc-based methods when interpreting diagnostic test results near the resistance breakpoint.

The CLSI susceptible breakpoint of 8 μg/mL for ampicillin susceptibility is likely conservative for lower UTIs in companion animals, as amoxicillin (ampicillin is used in vitro to predict amoxicillin susceptibility) achieves very high urinary concentrations—averaging > 200 μg/mL over a typical dosing interval for typical doses in both dogs and cats [16, 17]. In human medicine, amoxicillin has been shown to effectively treat lower UTIs caused by bacterial pathogens with MICs far exceeding this breakpoint, owing to these elevated urinary concentrations [18]. However, these high urinary concentrations are not applicable to upper UTIs, such as pyelonephritis, where tissue penetration is critical. Treatment of upper UTIs should be guided by AST using tissue-specific breakpoints to ensure adequate therapeutic concentrations at the site of infection.

The low prevalence of ampicillin-resistant isolates resulted in a small number of positive cases ($n = 19$), leading to a wide CI for sensitivity (66.86%–98.72%). Future studies with larger sample sizes or higher resistance prevalence could provide more precise estimates of the test's performance. Additionally, the test's reliance on visual turbidity assessment introduces subjectivity; future iterations could incorporate more objective turbidity assessments with a microbiology densitometer or simply by comparing to a set of standardized McFarland turbidity reference tubes to enhance reproducibility. Excluding patients with recent antibiotic exposure could further improve the test's reliability, as demonstrated by the probable confounding effect of recent cefovecin administration in Discrepant Result 1.

The broth turbidity test's simplicity—requiring only MH broth, ampicillin discs, sterile tubes, sterile inoculating loops, and a basic incubator—makes it accessible for in-house use, with very low material costs (e.g., under $5 per test). Compact low-cost incubators are widely available for purchase and could be used for other useful purposes in veterinary hospitals. For example, a broth without ampicillin could be inoculated to rule out uropathogens in cases such as cystitis in young, otherwise healthy cats. Traditional culture and susceptibility testing often exceed $100 per sample and can easily double the total cost of diagnosing and treating a UTI. In addition to the financial burden of the gold standard test, results typically require 48–72 h, delaying treatment decisions. By providing results within 16–24 h at a very low cost, the broth turbidity test enables practitioners to make timely decisions, potentially improving outcomes while encouraging the use of a first-line antimicrobial, a critical consideration in the context of antimicrobial stewardship [19].

5 | Limitations

This study has several limitations. First, the low prevalence of resistant bacteria limited the number of positive cases, affecting the precision of sensitivity estimates. Second, the test's performance in samples with low bacterial loads or recent antibiotic exposure suggests that modifications (e.g., larger inoculum volumes, exclusion of recently treated patients) might be needed to enhance sensitivity and reliability. Third, the study was conducted at a single diagnostic laboratory, and regional variations in resistance patterns might influence the test's generalizability. While the broth turbidity test detects ampicillin resistance, it does not provide organism identification or susceptibility

profiles for alternative antibiotics, necessitating follow-up testing for positive results. Finally, the study is limited by its in vitro design. Future studies should assess patient outcomes, along with in vitro susceptibility of bacterial isolates, to determine the usefulness of the diagnostic test more accurately.

6 | Conclusion

The ampicillin broth turbidity test offers a rapid, low-cost method for detecting ampicillin-resistant lower UTIs in dogs and cats, with high specificity and substantial agreement with gold standard traditional microbiological methods. Its perfect specificity in identifying ampicillin-susceptible infections underscores its reliability in avoiding unnecessary treatment escalation. However, its sensitivity might be affected by low bacterial loads, recent antibiotic administration, or variability in MIC and disc content near the resistance breakpoint. The CLSI breakpoint of 8 μg/mL for ampicillin susceptibility is likely conservative for lower UTIs, given that amoxicillin achieves urinary concentrations greatly exceeding 8 μg/mL. The test's clinical value could likely be maximized by limiting its use to infections with high CFUs, as identified by urinalysis, and excluding animals with recent antibiotic exposure.

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Disclosure

Authors declare no off-label use of antimicrobials.

Ethics Statement

Authors declare no institutional animal care and use committee or other approval was needed. Authors declare human ethics approval was not needed.

Conflicts of Interest

The authors declare no conflicts of interest.

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